The examination is being carried out on the **following application documents**

Description, Pages

1-25 filed with entry into the regional phase before the EPO

Claims, Numbers

1-11 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/19-19/19 filed with entry into the regional phase before the EPO

1 PRIOR ART

Reference is made to the following documents:

D1 JP 2022 043847 A (FUJI ELECTRIC CO LTD) 16 March 2022

D2 GB 1 531 844 A (SICK OPTIK ELEKTRONIK ERWIN) 8 November 1978

D3 JP 2002 258422 A (NIKON CORP) 11 September 2002

D4 JP 2008 261885 A (PERCEPTRONIX MEDICAL INC) 30 October 2008

D5 US 2015/247788 A1 (PAUL THOMAS ALFRED [CH] ET AL) 3 September 2015

D6 JP H05 257068 A (FUJITSU LTD.) 8 October 1993

D7 US 6 583 873 B1 (GONCHAROV ALEXANDER F [US] ET AL) 24 June 2003

2 NOVELTY AND INVENTIVE STEP

The present application does not meet the requirements of Article 52(1) EPC, because the subject-matter of **claims 1, 2, 5, 6, 11** is not new in the sense of Article 54(1) and (2) EPC and because the subject-matter of **claims 1-11** does not involve an inventive step in the sense of Article 56 EPC for the reasons presented below.

2.1 **Independent claim 1**

It is noted that in **claim 1**, the notch filter is only defined in terms of its function, such as having a restriction band for reducing an intensity at any peak of the radiation spectrum of the light emitted by the light source. Thus, the subject-matter of **claim 1** cannot be considered to be new in view of **D1** or **D2** for the following reasons.

2.1.1 **D1** [Fig. 1 and attached machine translation, par. 20-39] discloses a gas analyzer that measures a concentration of a measurement target component which is contained in a sample gas (par. 20), the gas analyzer comprising:
a light source unit that emits light having an absorption wavelength of the measurement target component (Fig. 1: light source unit 2);
a cell that houses one or more reflecting mirrors that reflect the light, in a space in which the sample gas is encapsulated (Fig. 1: cell 10, reflection mirrors 12-1, 12-2, 12-3);
a light receiving element that acquires a radiation spectrum of the light which has passed through the cell (Fig. 1: light receiving element 18 and par. 27);
a notch filter that is arranged in any optical path from the light source unit to the light receiving element and that has a restriction band for reducing an intensity at any peak of the radiation spectrum of the light which is emitted by the light source unit (Figs. 2-6 and par. 29-39; the coating of the mirrors acts as a notch filter by reducing the reflection and therefore the intensity of an emission peak of the light source); and
a processing unit that processes a light reception signal of the light receiving element to measure the concentration of the measurement target component (Fig. 1: signal processing circuit 20; par. 28).

Hence, the subject-matter of **claim 1** is not new in view of **D1**.

2.1.2 **D2** [Figs. 1, 2, 8; page 3, lines 64-83; page 4, line 105 - page 5, line 48; page 8, lines 57-62] discloses a gas analyzer that measures a concentration of a measurement target component which is contained in a sample gas (page 3, lines 64-83), the gas

analyzer comprising:
a light source unit that emits light having an absorption wavelength of the measurement target component (Fig. 1: light source unit 2);
a cell that houses one or more reflecting mirrors that reflect the light, in a space in which the sample gas is encapsulated (Fig. 1: mirror 13 which consists of triple mirrors, cell defined by features 76, 72, 14; page 4, lines 105-114 and page 5, lines 7-9);
a light receiving element that acquires a radiation spectrum of the light which has passed through the cell (Figs. 1, 8: photo-receiver 17 and filter wheel 18, which separately detect light of different wavelengths; cf. page 5, lines 23-48);
a notch filter that is arranged in any optical path from the light source unit to the light receiving element and that has a restriction band for reducing an intensity at any peak of the radiation spectrum of the light which is emitted by the light source unit (Figs. 1, 2: filter wheel 18; and page 5, lines 37-48; the filter stops certain emission peaks of the mercury-vapour lamp and therefore acts as a notch filter); and
a processing unit that processes a light reception signal of the light receiving element to measure the concentration of the measurement target component (page 8, lines 57-62: "computing stage 68").

Hence, the subject-matter of **claim 1** is not new in view of **D2**.

2.1.3 **NOTE:** Even if the notch filter would be clarified, such as to be a transmissive notch filter (based on the present application, Fig. 5 and par. 45), the subject-matter of claim 1 would still not be new in view of **D2**, which discloses a transmissive filter for blocking peak emission wavelengths of the light source (Figs. 1, 2: filter wheel 18).

In addition, such a claim would not be inventive when starting from **D1** as the closest prior art for the following reasons.

D1 already identifies the problem of undesired emission peaks and teaches to suppress them by using a specific reflective layer (Figs. 2-4 and translation, par. 29-35). Thus, the difference would be that instead of a specific mirror coating a transmissive notch filter is used to reduce the intensity of undesired emission peaks.

Technical effect would be to provide an alternative solution to reduce the effect of undesired emission peaks.

Thus, the objective technical problem would be be formulated as how to find an alternative way of suppressing the undesired emission peaks.

However, the use of transmissive notch filters for blocking certain undesired bands of a light source is generally known to the skilled person and is, for example, described in each of **D3** (Figs. 10, 11 and attached translation, par. 3-6), **D4** (attached translation, par. 16) or **D5** (par. 107).

Thus, it would be obvious to the skilled person to select a transmissive notch filter in order solve the objective technical problem formulated above, thereby arriving at the subject-matter of (clarified) claim 1 without exercising an inventive step.

2.2 **Dependent claims 2 - 11** do not appear to contain any additional features which, in combination with the features of any claim to which it refers, meet the requirements of the EPC with respect to novelty and/or inventive step.

-- Claim 2 [novelty]: see **D1**, Fig. 1 or **D2**, Fig. 1.

-- Claim 5 [novelty]: claim 5 does not define two mirrors, but only one second reflecting mirror. Thus, the mirror 13 of **D2**, Fig. 1 can be considered to represent a second mirror, which last reflects light through the emission window 20 to the receiver 17, wherein the notch filter 18 is arranged in the optical patch between the second mirror 13 and the receiver 17.

-- Claim 6 [novelty]: the actuator of the filter wheel 18 of **D2**, Fig. 1 may be considered to represent an adjustment unit.

-- Claim 11 [novelty]: each of the mirrors 12-1, 12-2 or 12-3 of **D1** can be a notch filter, see translation, par. 41-43.

-- Claims 3-5 [obviousness]: the position of the notch filter relates to an obvious choice in view of **D1** modified to comprise a transmissive notch filter as discussed under point 2.1.3 above

-- Claims 6-10 [obviousness]: it is generally known to the skilled person that the central wavelength of the notch filter depends on the incident angle (e.g. D6, Fig. 4 and attached translation, par. 17-20; or D7, Fig. 7). Thus it would be obvious to the skilled person to implement an incident angle adjustment unit to set/adjust the central wavelength of the notch filter of the modified device of **D1** (modified as discussed under point 2.1.3 above) and to correct the received signals in accordance with the set incident angle.

3 GENERAL OBJECTIONS

3.1 Independent claim 1 is not in the two-part form in accordance with Rule 43(1) EPC, which in the present case would be appropriate, with those features known in combination from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

3.2 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC).

3.3 To meet the requirements of Rule 42(1)(b) EPC, **D1 - D4** should be identified in the description and the relevant background art disclosed therein should be briefly discussed.